

Jetpack Compose Components & Libraries

Planned Jetpack Compose components and libraries for the future implementation of MASARAK, with clear justification and a dependency/compatibility plan. No production code is required at this milestone.

1. Planned Jetpack Compose Components

Ready-made Compose / Material 3 components planned for the current MASARAK screens.

COMPONENT	WHERE USED IN MASARAK	JUSTIFICATION
Scaffold	Main role screens	Provides the overall screen structure for the top bar, main content, bottom navigation, and Snackbar.
TopAppBar	Home, Chat, Profile, and detail screens	Provides a consistent header for screen titles, back navigation, notifications, and other actions.
NavigationBar + NavigationBarItem	Bottom navigation for all three roles	Supports the four main destinations for Student, Field Supervisor, and Academic Supervisor.
TabRow + Tab	Trainee and Student detail screens	Lets supervisors switch between Tasks, Attendance, and Weekly Evaluation without opening separate screens.
LazyColumn	Tasks, Students, Trainees, Notifications, and Chat lists	Displays scrollable lists efficiently when the amount of content can grow.
Row / Column / Box / Spacer	Across all layouts	Provides the basic building blocks for arranging content vertically, horizontally, in layers, and with consistent spacing.
Card	Task cards, weekly records, summaries, and feedback	Groups related information into clear visual sections that match the MASARAK interface style.
ListItem	Students, Trainees, Chat, and Notifications	Provides a consistent row structure for repeated list content.
Text / Icon / IconButton	All screens	Displays labels, values, navigation icons, status icons, and icon-only actions.
Button / OutlinedButton / TextButton	Submit, Approve, Return, Confirm, Cancel, and other actions	Separates primary, secondary, and low-emphasis actions clearly for the user.
OutlinedTextField	Login, Create Account, Add Task, Attendance, Weekly Grade, and Linking	Collects the text, numeric, password, multiline, and feedback input required by the app.

Component	Where Used in MASARAK	Justification
RadioButton	Role selection, Individual/Team, and relevance classification	Supports choosing one option from a small set of mutually exclusive choices.
LinearProgressIndicator	Student training-hours summary	Shows the student's confirmed training hours compared with the required total.
AlertDialog	Mark Task as Incomplete	Confirms an important action before it is applied.
ModalBottomSheet	Reset Password, Add Attachment, Link Trainee, Link Student	Handles short secondary workflows without creating an additional full screen.
Snackbar	Submission, confirmation, and error feedback	Shows brief feedback after an action without interrupting the user's current workflow.
DatePickerDialog + DatePicker	Add Task	Provides the standard Material date-selection interface for the task completion date.

2. Planned Jetpack Libraries

Core libraries planned for the future MASARAK implementation.

Jetpack Compose
Justification: Will be used to build the MASARAK Android user interface using declarative Kotlin UI.

Material 3
Justification: Provides the ready-made Material components needed for buttons, cards, text fields, navigation, dialogs, and other UI elements.

Navigation Compose
Justification: Will manage navigation between shared screens and the Student, Field Supervisor, and Academic Supervisor flows.

ViewModel
Justification: Will hold screen state and handle user events so UI logic remains separate from the Compose screens in the planned UDF architecture.

Additional libraries such as Room, WorkManager, Retrofit, or Firebase are not committed at this stage. They will only be added later if the implementation requires them.

3. Dependencies & Compatibility

The following dependencies are the main ones expected for the current plan. Exact version numbers will be selected during implementation after checking compatibility with the project setup.

DEPENDENCY	PURPOSE	COMPATIBILITY APPROACH
<code>androidx.compose.compose-bom</code>	Keeps Jetpack Compose library versions aligned.	Use one BOM version so Compose UI and Material-related libraries remain compatible with each other.
<code>androidx.compose.material3:material3</code>	Provides Material 3 UI components.	Use the version managed through the Compose BOM rather than selecting an unrelated version manually.
<code>androidx.navigation:navigation-compose</code>	Provides navigation between MASARAK screens.	Select a stable version that is compatible with the Android and Compose setup used by the project.
<code>androidx.lifecycle:lifecycle-viewmodel-compose</code>	Allows Compose screens to use ViewModels.	Use a Lifecycle version compatible with the project SDK and Compose environment.

Dependency versions are not fixed in Milestone 2 because production implementation has not started yet. The versions will be verified together before coding to avoid library conflicts.